

Name: _____

Answer all questions below.

1. Assume the producer runs the following code :

```
pthread_mutex_lock(&lock);  
pthread_cond_wait(&cond, &lock);  
pthread_mutex_unlock(&lock);
```

And the consumer runs :

```
pthread_mutex_lock(&lock);  
pthread_cond_signal(&cond);  
pthread_mutex_unlock(&lock);
```

Explain a case in which a deadlock will happen (e.g. a thread could sleep forever).

2. Assume there are zero bytes free. The following happen (in order)

- (a) Thread T1 wants to allocate 10 bytes
- (b) Thread T2 wants to allocate 75 bytes
- (c) Thread T3 wants to free 25 bytes

Write pseudocode for both allocate and free that will allow Thread T1 to allocate after Thread T3 frees.

3. When using conditional waits, why must `pthread_mutex_lock(&lock)` be called before `pthread_cond_wait(&cond, &lock)`?